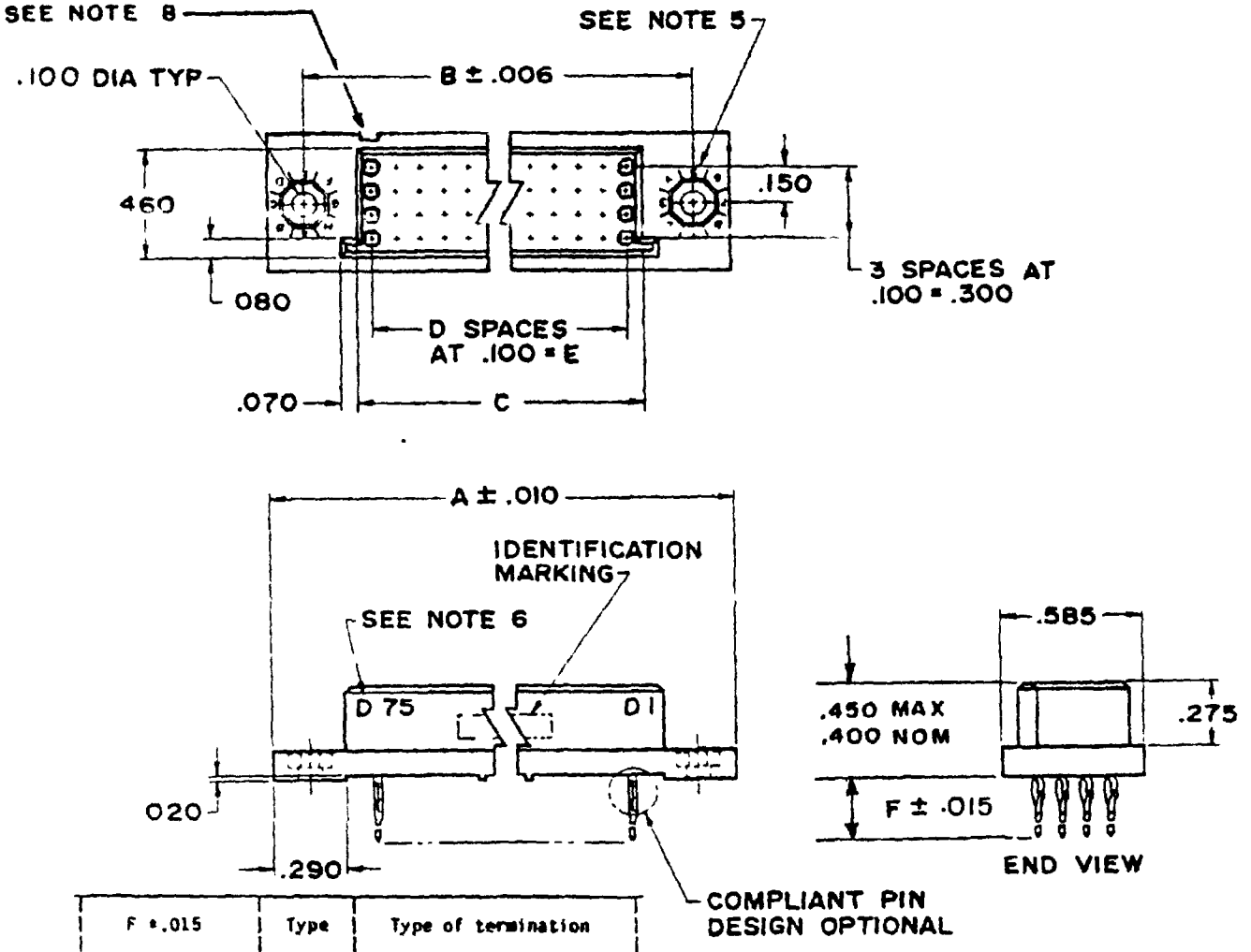
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES:
4-ROW RECEPTACLE ASSEMBLIES, 100 THROUGH 300 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



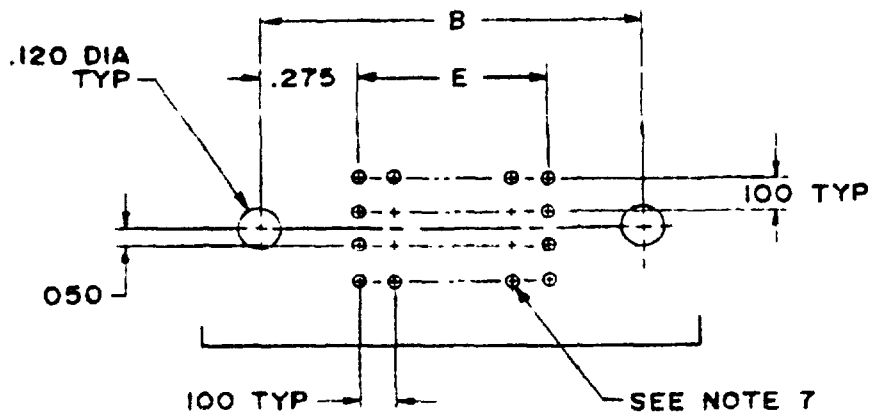
F ±.015		Type	Type of termination
Inches	mm		
.733	18 62	I	Compliant pin
.533	13 54	II	"
328	8.33	III	"
250	6.35	IV	"
.177	4.50	V	"
.190	4.83	VI	Solder post (see note)
.165	4.19	VII	"
.140	3.56	VIII	"
.120	3 05	IX	"

NOTE:
Solder post shall be a .036 (0 91 mm) maximum diameter or .036 (0 91 mm) maximum across the diagonal

FIGURE 1 Connectors, receptacle assemblies 100 - 54 mm spacing, 4-1 to right 3+ contacts

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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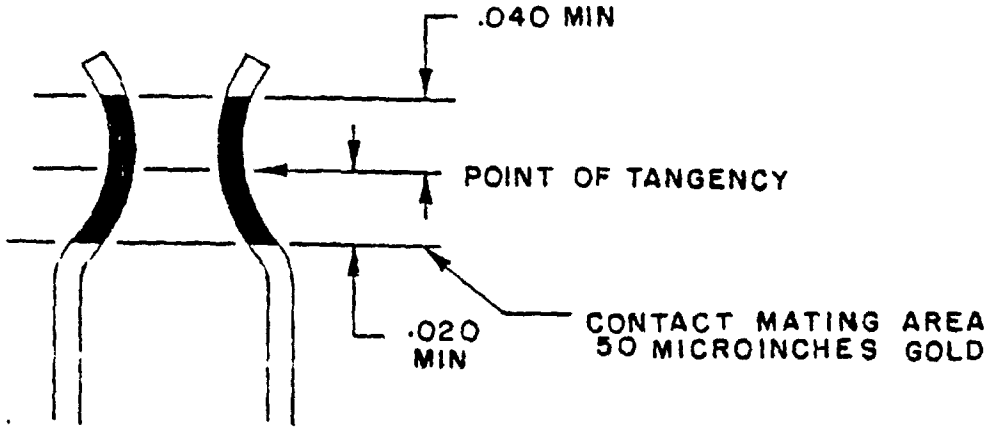
RECOMMENDED CIRCUIT LAYOUT

Inches	mm	Inches	mm
.006	0.15	.190	4.83
.015	0.38	.250	6.35
020	0.51	.275	6.98
050	1.27	.290	7.37
.070	1.78	.300	7 62
.080	2.03	.328	8.33
.100	2.54	.400	10.16
.115	2.92	460	11 68
.120	3.05	.533	13.54
.140	3.56	.550	13.97
.149	3.78	.585	14.86
.150	3.81	.733	18.62
.165	4.19		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4 These connectors mate with connectors specified in MIL-C-55302/179, MIL-C-55302/181, MIL-C-55302/188 and MIL-C-55302/189.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row closest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .037-.043 (0.94-1.09 mm). Holes for compliant pin shall be manufactured as follows: .0453±.0010 (1.15±0.0254 mm) drilled hole (1.15 mm drill) .001-.003 (0.025-0.076 mm) thick copper on wall, .0003 (0.0076 mm) minimum tin lead plating, finished hole diameter .037-.043 (0.94-1.09 mm) after plating, .036-.043 (0 91-1.09 mm) after reflow, copper hardness (Knoop) 150 maximum.
8. Optional notches of .050 maximum depth (typical shown) shall have .115 material minimum between them

FIGURE 1 Connectors, receptacle assemblies .100 (2.54 mm) spacing 100 through 300 contacts Continued.

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DESIGN OF TINES OPTIONAL

Inches	mm
.020	0.51
.040	1.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Contact plating

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TABLE I Dash number, number of contacts, contact type, and dimensions

Dash number	Number of contacts	Contact type	A	B	C	D	E	Contact identification number
01 02 03 04 05 06 07 08 09	100	I II III IV V VI VII VIII IX	3.250 (82.55)	2.950 (74.93)	2.624 (64.11)	24	2.400 (60.96)	1-25
10 11 12 13 14 15 16 17 18	120	I II III IV V VI VII VIII IX	3.750 (95.25)	3.450 (87.63)	3.024 (76.81)	29	2.900 (73.66)	1-30
19 20 21 22 23 24 25 26 27	128	I II III IV V VI VII VIII IX	3.950 (100.33)	3.650 (92.71)	3.224 (81.89)	31	3.100 (78.74)	1-32
28 29 30 31 32 33 34 35 36	140	I II III IV V VI VII VIII IX	4.250 (107.95)	3.950 (100.33)	3.524 (89.51)	34	3.400 (86.36)	1-35
37 38 39 40 41 42 43 44 45	160	I II III IV V VI VII VIII IX	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)	1-40

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash number	Number of contacts	Contact type	A	B	C	D	E	Contact identification number
46 47 48 49 50 51 52 53 54	180	I II III IV V VI VII VIII IX	5.250 (133.35)	4.950 (125.73)	4.524 (114.91)	44	4.400 (111.76)	1-45
55 56 57 58 59 60 61 62 63	200	I II III IV V VI VII VIII IX	5.750 (146.05)	5.450 (138.43)	5.024 (126.61)	49	4.900 (124.46)	1-50
64 65 66 67 68 69 70 71 72	220	I II III IV V VI VII VIII IX	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)	1-55
73 74 75 76 77 78 79 80 81	240	I II III IV V VI VII VIII IX	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)	1-60
82 83 84 85 86 87 88 89 90	260	I II III IV V VI VII VIII IX	7.250 (184.15)	6.950 (176.53)	6.524 (165.71)	64	6.400 (162.56)	1-65

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TABLE I. Dash number, number of contacts, contact type and dimensions - Continued

Dash number	Number of contacts	Contact type	A	B	C	D	E	Contact identification number
91 92 93 94 95 96 97 98 99	266	I II III IV V VI VII VIII IX	7.450 (189.23)	7.150 (181.61)	6.724 (170.79)	66	6.600 (167.64)	1-67
100 101 102 103 104 105 106 107 108	280	I II III IV V VI VII VIII IX	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)	1-70
109 110 111 112 113 114 115 116 117	300	I II III IV V VI VII VIII IX	8.250 (209.55)	7.950 (201.93)	7.524 (191.11)	74	7.400 (187.96)	1-75

TABLE II. Contact plating requirements.

Plating style	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area shall have a minimum plating (see figure 2) in accordance with MIL-C-55302. Compliant pin area (contact type I, .748 inch minimum length; type II, .548 minimum length, type III, .343 inch minimum length, type IV, .265 inch minimum length; type V, .192 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302. Solder post area (contact type VI, .205 minimum length; type VII, .190 inch minimum length, type VIII, .165 inch minimum length, type IX, .135 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.
C	Overall contact finish: Shall be palladium or palladium nickel alloy (80/14 ±6 percent), 50 microinches minimum thickness, with a gold flash of 3 microinches minimum, over a nickel underplate of 50 microinches minimum in accordance with QQ-N-290, and shall meet the requirements of MIL-P-45209.

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REQUIREMENTS.

Design and construction

Dimensions and configuration. See figure 1 and table I.

Material: In accordance with MIL-C-55302.

Plating: See figure 2 and table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, part number MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-C-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or -16. Jackscrews are purchased separately.

Compliant section shall meet the requirements of MIL-STD-2166, type III. For use with .093 minimum thick printed wiring boards. Sequence of tests and number of samples shall be in accordance with table III of MIL-STD-2166.

Wire wrappost: See MIL-STD-1130.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance. No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part number. M55302/185 (and dash number from table I and plating style from table II).

Example: M55302/185-01A.

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Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 11, 19, 80

Preparing activity
Army - CR

Agent
DLA - ES

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